CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 12: Surface Areas and Volumes

Step-by-step solutions to the textbook exercises: 17 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 12.1

Q1 Two cubes, each of volume 64 cm^3, are joined end to end. Find the surface area of the cuboid formed. (Take $\pi = \frac{22}{7}$ in this chapter unless told otherwise.)

SOLUTION

Side of each cube $= \sqrt[3]{64} = 4$ cm. The cuboid measures $8 \times 4 \times 4$ cm.

Surface area $= 2(lb + bh + hl) = 2(32 + 16 + 32) = 160$.

Answer: 160 cm^2

Q2 A vessel is a hollow hemisphere with a hollow cylinder mounted on it. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find its inner surface area.

SOLUTION

$r = 7$ cm, and the height of the cylinder is $13 - 7 = 6$ cm.

Inner surface area $= 2\pi r^2 + 2\pi rh = 2\pi r(r + h) = 2 \times \frac{22}{7} \times 7 \times 13 = 572$.

Answer: 572 cm^2

Q3 A toy is a cone of radius 3.5 cm mounted on a hemisphere of the same radius. Its total height is 15.5 cm. Find the total surface area of the toy.

SOLUTION

Height of the cone $= 15.5 - 3.5 = 12$ cm, so its slant height is $l = \sqrt{3.5^2 + 12^2} = \sqrt{156.25} = 12.5$ cm.

Total surface area $= \pi rl + 2\pi r^2 = \pi r(l + 2r) = \frac{22}{7} \times 3.5 \times (12.5 + 7) = 11 \times 19.5 = 214.5$.

Answer: 214.5 cm^2

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Q4 A cube of side 7 cm has a hemisphere placed on top of it. What is the largest possible diameter of the hemisphere? Find the total surface area of the combined solid.

SOLUTION

The hemisphere must fit on one face of the cube, so its greatest diameter is 7 cm and $r = 3.5$ cm.

Surface area = surface of the cube − area of the circle covered + curved surface of the hemisphere $= 6 \times 49 - \pi r^2 + 2\pi r^2 = 294 + \pi r^2$.

$= 294 + \frac{22}{7} \times 3.5 \times 3.5 = 294 + 38.5 = 332.5$.

Answer: greatest diameter 7 cm, surface area 332.5 cm^2

Q5 A hemispherical depression is cut out of one face of a cubical wooden block, with the diameter l of the hemisphere equal to the edge of the cube. Find the surface area of the remaining solid.

SOLUTION

The radius of the hemisphere is $\frac{l}{2}$. One face loses a circle of area $\pi\left(\frac{l}{2}\right)^2$ and gains the curved surface $2\pi\left(\frac{l}{2}\right)^2$.

Surface area $= 6l^2 - \frac{\pi l^2}{4} + \frac{\pi l^2}{2} = 6l^2 + \frac{\pi l^2}{4}$.

Answer: $\frac{l^2}{4}(24 + \pi)$ square units

Q6 A medicine capsule is a cylinder with a hemisphere at each end. The whole capsule is 14 mm long and 5 mm in diameter. Find its surface area.

SOLUTION

$r = 2.5$ mm. The cylindrical part is $14 - 2 \times 2.5 = 9$ mm long.

Surface area $= 2\pi rh + 2 \times 2\pi r^2 = 2\pi r(h + 2r) = 2 \times \frac{22}{7} \times 2.5 \times 14 = 220$.

Answer: 220 mm^2

Q7 A tent is a cylinder with a conical top. The cylindrical part is 2.1 m high and 4 m in diameter, and the slant height of the top is 2.8 m. Find the area of canvas used, and its cost at ₹500 per m^2. (The base is not covered.)

SOLUTION

$r = 2$ m. Canvas area = curved surface of the cylinder + curved surface of the cone $= 2\pi rh + \pi rl = \pi r(2h + l)$.

$= \frac{22}{7} \times 2 \times (4.2 + 2.8) = 44$ m^2.

Cost $= 44 \times 500 =$ ₹22,000.

Answer: 44 m^2 and ₹22,000

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Q8 A solid cylinder of height 2.4 cm and diameter 1.4 cm has a cone of the same height and diameter carved out of it. Find the total surface area of what is left, to the nearest cm^2.

SOLUTION

$r = 0.7$ cm, $h = 2.4$ cm, and the slant height of the cone is $l = \sqrt{0.7^2 + 2.4^2} = \sqrt{6.25} = 2.5$ cm.

Total surface area = curved surface of the cylinder + one circular base + curved surface of the cone $= 2\pi rh + \pi r^2 + \pi rl = \pi r(2h + r + l)$.

$= \frac{22}{7} \times 0.7 \times (4.8 + 0.7 + 2.5) = 2.2 \times 8 = 17.6\ \text{cm}^2$.

Answer: about 18 cm^2

Q9 A wooden article is made by scooping out a hemisphere from each end of a solid cylinder of height 10 cm and base radius 3.5 cm. Find its total surface area.

SOLUTION

Total surface area = curved surface of the cylinder + two hemispherical surfaces $= 2\pi rh + 2 \times 2\pi r^2 = 2\pi r(h + 2r)$.

$= 2 \times \frac{22}{7} \times 3.5 \times (10 + 7) = 22 \times 17 = 374$.

Answer: 374 cm^2

2 Exercise 12.2

Q1 A solid is a cone standing on a hemisphere, both of radius 1 cm, and the height of the cone equals its radius. Find the volume of the solid in terms of π.

SOLUTION

Volume $= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = \frac{2}{3}\pi(1)^3 + \frac{1}{3}\pi(1)^2(1) = \pi$.

Answer: π cm^3

Q2 A model is a cylinder with a cone attached at each end, made of thin aluminium sheet. Its diameter is 3 cm, its length is 12 cm and each cone is 2 cm high. Find the volume of air in the model.

SOLUTION

$r = 1.5$ cm. The cylinder is $12 - 2 - 2 = 8$ cm long.

Volume $= \pi r^2 \times 8 + 2 \times \frac{1}{3}\pi r^2 \times 2 = \pi \times 2.25 \times \left(8 + \frac{4}{3}\right) = 21\pi = 21 \times \frac{22}{7} = 66$.

Answer: 66 cm^3

Q3 A gulab jamun contains sugar syrup up to about 30% of its volume. Each is shaped like a cylinder with two hemispherical ends, with length 5 cm and diameter 2.8 cm. About how much syrup is there in 45 gulab jamuns?

SOLUTION

$r = 1.4$ cm. The cylindrical part is $5 - 2.8 = 2.2$ cm long.

Volume of one $= \pi r^2 h + \frac{4}{3}\pi r^3 = \pi r^2 \left(h + \frac{4r}{3}\right) = \frac{22}{7} \times 1.96 \times \left(2.2 + \frac{5.6}{3}\right) \approx 25.05$ cm^3.

Volume of 45 ≈ 1127.28 cm^3. Syrup $= 30\%$ of this ≈ 338 cm^3.

Answer: about 338 cm^3

Q4 A wooden pen stand is a cuboid 15 cm by 10 cm by 3.5 cm with four conical depressions, each of radius 0.5 cm and depth 1.4 cm. Find the volume of wood in the stand.

SOLUTION

Volume of the cuboid $= 15 \times 10 \times 3.5 = 525$ cm^3.

Volume of four cones $= 4 \times \frac{1}{3} \times \frac{22}{7} \times 0.5^2 \times 1.4 = \frac{22}{15} \approx 1.47$ cm^3.

Answer: $525 - 1.47 = 523.53$ **cm^3**

Q5 A vessel is an inverted cone of height 8 cm with an open top of radius 5 cm, full of water. Lead shots, each a sphere of radius 0.5 cm, are dropped in and one-fourth of the water flows out. Find the number of lead shots.

SOLUTION

Volume of water $= \frac{1}{3}\pi \times 5^2 \times 8 = \frac{200\pi}{3}$ cm^3. The water that flows out is $\frac{1}{4}$ of this, $\frac{50\pi}{3}$ cm^3, and equals the volume of the shots.

Volume of one shot $= \frac{4}{3}\pi \times 0.5^3 = \frac{\pi}{6}$ cm^3.

Number of shots $= \frac{50\pi}{3} \div \frac{\pi}{6} = 100$.

Answer: 100

Q6 A solid iron pole is a cylinder of height 220 cm and base diameter 24 cm, with another cylinder of height 60 cm and radius 8 cm on top. Find the mass of the pole if 1 cm^3 of iron has a mass of about 8 g. (Use $\pi = 3.14$.)

SOLUTION

Volume $= \pi \times 12^2 \times 220 + \pi \times 8^2 \times 60 = 3.14 \times (31680 + 3840) = 3.14 \times 35520 = 111532.8$ cm^3.

Mass $= 111532.8 \times 8 = 892262.4$ g.

Answer: about 892.26 kg

Q7 A solid made of a cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is placed upright in a cylinder full of water, touching the bottom. The cylinder has radius 60 cm and height 180 cm. Find the volume of water left in the cylinder.

SOLUTION

Volume of the cylinder $= \pi \times 60^2 \times 180 = 648000\pi$ cm^3.

Volume of the solid $= \frac{1}{3}\pi \times 60^2 \times 120 + \frac{2}{3}\pi \times 60^3 = 144000\pi + 144000\pi = 288000\pi$ cm^3.

Water left $= 648000\pi - 288000\pi = 360000\pi = 360000 \times \frac{22}{7} \approx 1131428.57$ cm^3.

Answer: about 1131428.57 cm^3, that is about 1.131 m^3

Q8 A spherical glass vessel has a cylindrical neck 8 cm long and 2 cm in diameter, and the spherical part is 8.5 cm in diameter. A child measures its volume as 345 cm^3. Is she correct? (Take these as inside measurements and $\pi = 3.14$.)

SOLUTION

Volume of the sphere $= \frac{4}{3} \times 3.14 \times 4.25^3 = \frac{4}{3} \times 3.14 \times 76.765625 \approx 321.39$ cm^3.

Volume of the neck $= 3.14 \times 1^2 \times 8 = 25.12$ cm^3.

Total $\approx 321.39 + 25.12 = 346.51$ cm^3.

Answer: she is not correct. The volume is about 346.51 cm^3.

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